

Arnav Revankar

Junior Computer Engineer @ NJIT

github.com/avr34 | avr33@njit.edu | linkedin.com/in/avr33

EDUCATION

New Jersey Institute of Technology (Sept 2023 - Present)

Cumulative GPA: 3.65

BS Computer Engineering, Minor in Data Analytics

Newark, NJ

- **Coursework:** Circuits and Systems 1 & 2, Microprocessors, Computer Architecture, Applied Machine Learning, Digital Design, and others.
- **Activities:** IEEE-HKN Inductee, 2026 Hackathon Volunteer

County College of Morris (Sept 2022 - May 2023)

Cumulative GPA: 3.83

Challenger Program for High School Students

Randolph, NJ

- Completed over two semesters as a part-time student.

EXPERIENCE

Software Engineering Intern

March 2025 - August 2025

Mathnasium

Chatham, NJ

- Spearheaded a project developing a full-stack Javascript/Node.js email generator for parents of students at a Mathnasium tutoring center.
- Effectively employs a BNF Grammar, that is used to generate syntactically valid text from a set of customizable templates.
- Currently being used to send 110-150 emails per day, saving 5 centers 30-40 hours a week.

SKILLS

- **Languages:** C, C++, MATLAB, Java, Python, Go, SQL (MySQL), GNU Octave, $\text{\LaTeX}$, Javascript, VHDL
- **Software Tools:** Linux/Unix, KiCAD, FreeCAD, STM32CubeMX & IDE, PlatformIO, ESP-IDF, Git, Docker, Excel
- **Build Tools/Debuggers:** Apache Maven, Apache Ant, CMake, Make, MSVC & GCC, GDB, Valgrind
- **Libraries:** STM32 HAL, FreeRTOS, NumPy, PyTorch, Scikit Learn, Matplotlib, Gnuplot
- **Communication Protocols:** UART, I2C, SPI, CAN, USB 2.0, TCP and Socket programming in C/C++ and Go

PROJECTS

BitSum Lecture Summarizer ([GitHub](#))

Oct 2025 - Dec 2025

- Built a lecture summarizer tool, integrating Microsoft BitNet, OpenAI Whisper, and a custom feedforward network.
- Transcribes and summarizes lectures on CPU, 2x faster than human comprehension on average.
- Integrated with Docker to ensure reproducibility.

8-Channel Logic Analyzer ([Github](#))

Feb 2026 - May 2025

- USB Logic Analyzer built around an STM32F446, capable of sniffing upto 500kHz signals.
- DMA data acquisition, and run-length encoded data transmission, to optimize byte usage on USB 2.0 FS.
- UART, SPI, I2C, and CAN signal decoding via a custom Go backend
- GUI with PyQt5 in Python

SolarVision Autonomous Mower ([Github](#))

Sept 2025 - May 2025

- YOLO and LiDAR-based autonomous lawnmower, engineering capstone project. Still in progress
- Designed a custom 6-layer PCB for STM32N6 MCU, with external SPI NOR Flash and SDRAM Memory through the FMC.
- Trained a YOLOv8 Segmentation model on a custom dataset, and exported to C for the STM32N6 NPU, with the ST Edge AI Tool Suite.
- LiDAR data acquisition and path planning on an ESP32S3

ESP32 2nd Stage Bootloader

May 2026 - Present

- Building a toy 2nd stage bootloader for an ESP32S3 microcontroller.
- Planning on supporting basic peripheral initialization, and OTA firmware updates if possible.
- Written in C using ESP-IDF HAL library and tooling, as well as Doxygen for code documentation.